

COURSE PROGRAM

Generative AI for End-to-End Drug Discovery & Life Sciences

LEARN HOW AI IS DESIGNING TOMORROW'S MEDICINES TODAY

NETWORKING WITH EXPERTS & PEERS

HANDS-ON LEARNING APPROACH

REAL -WORLD AI USE CASES

INDUSTRY-RELEVANT CURRICULUM

CROSS-DOMAIN EXPERTISE (AI + LIFE SCIENCES)

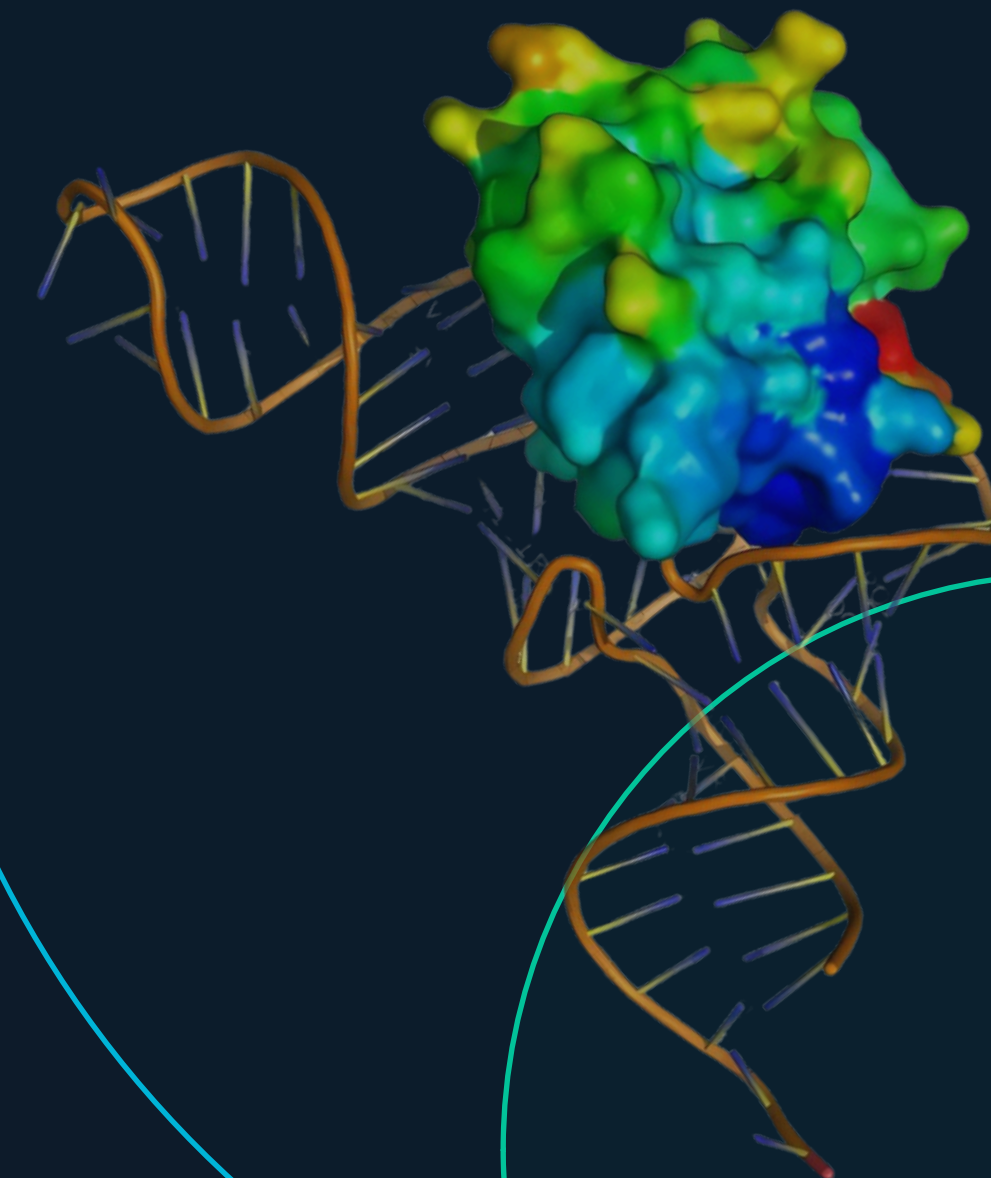
A comprehensive learning journey from AI foundations to enterprise-grade drug discovery systems

Beginner & Competent

Proficient

Expert

For More Information, visit our website: For More Information, visit our website: <https://genesysquantis.com>
or reach out to us at: connect@genquantis.com, ph no: 7036955133



Who Is This Program For?

A multidisciplinary program bridging AI, biology, and drug discovery for the next generation of innovators.



B.Pharm & M.Pharm

Students & professionals in pharmaceutical sciences



B.Tech Biotechnology

Engineering students specialising in biological sciences



B.Sc Life Sciences

Biotechnology & life sciences graduates



AI / ML Engineers

Data scientists transitioning into pharma & life sciences



Pharma Researchers

R&D professionals in drug discovery & development



Healthcare Professionals

Clinicians & translational medicine researchers (MBBS, MD Pharmacology)

Three Learning Tracks

Choose your entry point — every track leads to a deeper understanding of AI in drug discovery

01  Beginner & Competent		02  Proficient		03  Expert	
5 Days Duration		20 Sessions Total Sessions		12 Modules Structured Modules	
72 Topics In-depth Topics		12 Modules Expert Modules		12 Capstones Build-and-Deploy Projects	
Solid AI/ML foundations across the full drug discovery pipeline — from multi-omics data to clinical AI. ◦ AI & ML fundamentals for biology ◦ Multi-omics & target discovery ◦ Generative AI in chemistry ◦ Simulation & clinical AI ◦ AI systems & automation		Applied knowledge across specialised domains — protein design, autonomous labs, and regulatory AI. ◦ Multi-omics & biomarker discovery ◦ Protein & antibody AI design ◦ Autonomous laboratory systems ◦ Generative chemistry & synthesis ◦ Clinical AI & regulatory compliance		Design and deploy production-grade AI systems — from biological foundation models to enterprise pharma platforms. ◦ Biological foundation models (BioFM) ◦ Closed-loop autonomous discovery ◦ Molecular diffusion models ◦ Clinical digital twin platforms ◦ Enterprise bio-AI infrastructure	

TRACK 01 · BEGINNER & COMPETENT

Beginner & Competent

5-Day Foundational Program · 20 Sessions

5-Day Program at a Glance

Each day builds upon the last — from AI foundations to full pipeline integration

 Day 1	 Day 2	 Day 3	 Day 4	 Day 5
Foundations → Data → Target Discovery	Protein Design → Autonomous Labs	Generative AI in Chemistry	Simulation → Safety → Clinical AI	AI Systems → Automation → Integration
Role of AI in Pharma & Biotech	Protein Structure Prediction	Molecular Representation & GNNs	AI in Molecular Dynamics	AI Research Assistants & NLP
Multi-omics Data Integration	Transformers in Biology (LLMs)	Generative Models (VAE / GAN / Diffusion)	AI + Generative Chemistry	Multi-Agent AI Systems
Supervised & Deep Learning	Generative AI for Protein Design	Hit ID & Lead Optimization	ADMET & Toxicity Prediction	AI Infrastructure & MLOps
AI-driven Target Identification	AI-driven Lab Automation	Retrosynthesis with AI	AI in Clinical Trials	End-to-End Pipeline Design

Days 1 & 2 — AI Foundations & Protein Design



Day 1 · Foundations → Data → AI for Target Discovery

Session 1: Role of AI in Pharma

- AI across the drug discovery pipeline
- ML vs DL vs Generative AI in life sciences

Session 2: Multi-omics Data

- Genomics, proteomics & clinical data
- AI for data integration, feature engineering & embeddings

Session 3: ML & Deep Learning

- Supervised vs unsupervised learning in biology
- CNNs, RNNs & representation learning for sequences

Session 4: AI-driven Target ID

- Multi-omics fusion using deep learning
- Biomarker discovery using ML models



Day 2 · AI for Protein Design → Autonomous Labs

Session 1: Protein Structure

- Protein structure prediction using AI
- Sequence-to-structure learning & embeddings

Session 2: Transformers in Biology

- LLMs for proteins, attention mechanisms
- Binding affinity prediction using DL

Session 3: Generative Protein AI

- De novo protein design & antibody design with AI
- Reinforcement learning in protein optimisation

Session 4: Autonomous Labs

- AI-driven lab automation systems & active learning
- Closed-loop self-learning experimentation

Days 3 & 4 — Generative Chemistry & Simulation AI



Day 3 · Generative AI in Chemistry → Molecular Engineering

Session 1: Molecular Representation

- SMILES, graphs & chemical embeddings
- Graph Neural Networks (GNNs) for chemistry

Session 2: Generative Models

- VAEs, GANs & Diffusion models for molecules
- AI-driven molecule generation & chemical space exploration

Session 3: Hit ID & Lead Optimisation

- Multi-objective optimisation using ML
- Property prediction models

Session 4: Retrosynthesis

- Reaction prediction using deep learning
- Neural symbolic methods in chemistry



Day 4 · Simulation AI → Safety AI → Clinical AI

Session 1: Molecular Dynamics

- AI in molecular dynamics simulations
- ML-based force fields & protein-ligand interaction prediction

Session 2: Generative Chemistry

- AI + Generative Chemistry hybrid models
- Stability prediction using ML & surrogate modelling

Session 3: ADMET & Toxicity

- ADMET prediction using ML/DL
- Explainable AI (XAI) in drug safety

Session 4: Clinical AI

- AI in clinical trials & predictive modelling
- Patient stratification & precision medicine using AI

Day 5 — AI Systems, Automation & Integration



Day 5 · AI Systems → Automation → Integration

Session 1: AI Research Assistants

- LLMs for scientific literature mining
- NLP & knowledge graphs in life sciences

Session 2: Multi-Agent AI Systems

- Autonomous decision-making pipelines
- AI orchestration in laboratory workflows

Session 3: AI Infrastructure

- MLOps & data pipelines for pharma
- Vector databases for biological data & scalable AI

Session 4: End-to-End Pipeline

- Designing a full AI drug discovery pipeline
- Multi-model workflows & capstone architecture design

Program Outcomes



Apply AI/ML across the full drug discovery pipeline



Design generative models for molecules & proteins



Build & operate autonomous lab AI systems



Simulate molecular interactions using AI & Generative methods



Integrate multi-model AI workflows for enterprise use

TRACK 02 · PROFICIENT

Proficient

45 Days · 12-Module Applied Skills Program · 72 In-Depth Topics

Proficient — 12-Module Curriculum

A structured progression through AI applications across the full drug discovery pipeline



Proficient — Modules 01 to 06

01 AI Foundations in Life Sciences

- Role of AI in pharma & biotech
- Drug discovery pipeline overview
- Genomics, proteomics & clinical data types
- ML & DL basics for biology
- Visualisation in life sciences

02 Multi-Omics & Target Discovery

- Multi-omics data integration
- AI-driven target identification
- Feature engineering in omics
- Biomarker discovery using ML
- Therapeutic target prioritisation

03 Protein Design & Bioengineering AI

- Protein structure prediction basics
- Transformer models in biology
- De novo protein design with AI
- Antibody optimisation using AI
- Binding affinity prediction

04 Autonomous Labs & Experimentation

- Lab automation concepts & robotics
- AI-driven experiment design
- Experiment tracking systems
- Real-time data feedback loops
- Autonomous discovery loops

05 Biological Foundation Models

- Introduction to LLMs for biology
- Aging clock & cellular models
- Clinical + omics integration
- Multispecies modelling
- Systems biology AI

06 Clinical & Translational AI

- Clinical trial phases & AI
- Predictive trial modelling
- Patient stratification methods
- AI in precision medicine
- Regulatory AI considerations

Proficient — Modules 07 to 12

07

Generative Chemistry & Drug Design

- Chemical space exploration basics
- Generative models — Diffusion & VAEs
- Hit identification workflows
- Scaffold hopping strategies
- Lead optimisation with ML

08

Synthetic Planning & Molecular Engineering

- Retrosynthesis basics
- Reaction prediction models
- Yield optimisation using AI
- Neural architectures in chemistry
- Industrial chemistry applications

09

Molecular Simulation & Biology

- Molecular dynamics basics
- Protein-ligand interaction modelling
- Generative chemistry basics
- RBFE binding energy calculations
- Stability analysis with AI

10

Drug Safety & Chemical Space Intelligence

- ADMET prediction methods
- Toxicity modelling with ML/DL
- Chemical space exploration
- SAR analysis frameworks
- Late-stage failure reduction

11

Scientific Research Automation

- AI research assistants
- Literature mining with NLP
- Multi-agent AI in research
- Lab protocol automation
- Conversational AI in science

12

Enterprise AI Infrastructure & Capstone

- AI infrastructure in pharma
- Data pipelines & vector databases
- Secure data architecture
- Scaling AI systems
- Capstone project guidance

TRACK 03 · EXPERT

Expert

90 Days · 12 Capstone Modules · Build Production-Grade AI Systems

Expert — Capstone Modules 01 to 06

01 Multi-Omics Foundation Model for Target Discovery CAPSTONE Multi-Omics Foundation Model for Target Discovery Tech Stack Transformers · GNNs · FAISS / Weaviate Predict novel therapeutic targets & biomarker discovery engine	02 AI-Powered Target Prioritisation Engine (Causal AI) CAPSTONE AI-Powered Target Prioritisation Engine (Causal AI) Tech Stack DoWhy / Pyro · Bayesian Networks · Causal SCMs Ranked targets with causal confidence scores	03 Protein LLM for De Novo Therapeutic Design CAPSTONE Protein LLM for De Novo Therapeutic Design Tech Stack Protein LLM (ESM/ProtGPT) · RLHF · Wet Lab Loop AI-designed antibodies & therapeutic enzymes
04 Closed-Loop Autonomous Drug Discovery Lab CAPSTONE Closed-Loop Autonomous Drug Discovery Lab Tech Stack LangGraph / AutoGen · Multi-Agent · RL Fully autonomous hypothesis testing pipeline	05 Generative Chemistry Engine with Diffusion Models CAPSTONE Generative Chemistry Engine with Diffusion Models Tech Stack Diffusion Models · RDKit · PyTorch Geometric Novel compounds with optimised ADMET properties	06 End-to-End AI Drug Discovery Pipeline (Enterprise) CAPSTONE End-to-End AI Drug Discovery Pipeline (Enterprise) Tech Stack Microservices · MLOps · Data Lake · Feature Store Production-ready full-stack pharma AI platform

Expert — Capstone Modules 07 to 12

07 AI-DRIVEN RETROSYNTHESIS & REACTION PLANNING ENGINE CAPSTONE AI-Driven Retrosynthesis & Reaction Planning Engine Tech Stack Transformer Reaction Prediction · Monte Carlo Tree Search Industrial-scale synthesis planning system	08 HYBRID AI + GENERATIVE CHEMISTRY SIMULATION ENGINE CAPSTONE Hybrid AI + Generative Chemistry Simulation Engine Tech Stack Schrodinger / OpenMM · ML Surrogate · Neural Operators 10-100x faster binding energy calculations	09 AI-BASED ADMET & TOXICITY EXPLAINABILITY PLATFORM CAPSTONE AI-Based ADMET & Toxicity Explainability Platform Tech Stack XAI (SHAP, LIME) · Graph Neural Networks Regulatory-ready AI drug safety platform
10 CLINICAL TRIAL DIGITAL TWIN PLATFORM CAPSTONE Clinical Trial Digital Twin Platform Tech Stack Digital Twin Modelling · Transformer Prediction Models Personalised virtual trial simulation system	11 SCIENTIFIC MULTI-AGENT RESEARCH COPILOT CAPSTONE Scientific Multi-Agent Research Copilot Tech Stack Multi-Agent (Researcher, Critic, Lab Planner) · LangGraph Autonomous research assistant for pharma	12 ENTERPRISE BIO-AI INFRASTRUCTURE & KNOWLEDGE CAPSTONE Enterprise Bio-AI Infrastructure & Knowledge Graph Tech Stack Neo4j / TigerGraph · Vector + Graph Hybrid Retrieval Decision intelligence system for pharma

Your Journey in AI-Powered Drug Discovery Begins Here.

From foundational AI concepts to enterprise-grade drug discovery platforms — this program equips you to lead the next generation of life sciences innovation.

5

Days · Beginner Program

45

Days · Proficient Track Program

90

Days · Expert Capstone Projects

Ideal for: B.Pharm · M.Pharm · B.Tech Biotechnology · B.Sc Life Sciences · AI/ML Engineers · Pharma Researchers · Healthcare Professionals (MBBS, MD Pharmacology)

For More Information, visit our website: For More Information, visit our website: <https://genesysquantis.com>
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